

[Request] SAM ALTMAN: “People talk about how much energy it takes to train an AI model ... But it also takes a lot of energy to train a human. It takes like 20 years of life and all of the food you eat during that time before you get smart.” how true is this comparison to a typical western lifestyle

Posted by u/so_how_can_i_help • 0 points • 30 comments

Tuypical western diet, lifestyle etc.

Comments

u/jon_muselee • 1 points • 2026-02-22 11:07 UTC

I'm more worried about how much energy it takes to just ask an AI one question. Compared to a human AI takes 100-10000 times more energy for that.

u/[deleted] • 1 points • 2026-02-22 07:54 UTC

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u/Affial • 1 points • 2026-02-22 07:48 UTC

Lmao, is it the first time he makes the comparison?
I knew the strongest of tech bros had this line of thoughts.

Yeah, we made it: a life it's on the market now (well, after slavery), a product with an energy class and everything.

Again: this is not progress. It's a *smart psychosis*.

u/Former_Print7043 • 15 points • 2026-02-22 06:53 UTC

Some of these tech billionaires need a leash. I am okay with the system making kooky savant types rich but to allow them to dictate policy to the lives of more perfectly average folks is just as silly as letting kings do it.

u/Unfair-Potato-5868 • 1 points • 2026-02-23 13:44 UTC

that's why we shouldn't allow them to get rich. wealth is power

u/[deleted] • 1 points • 2026-02-22 06:25 UTC

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u/1F61C • 5 points • 2026-02-22 06:16 UTC

In 2023, the total primary energy consumption per person in the United States was approximately 279 million British thermal units (MMBtu). This figure represents the total energy footprint, including transportation, home heating, and industrial goods, not just electricity consumption. That's 81766kWh. [link](https://www.eia.gov/tools/faqs/faq.php?id=85&t=1#:~:text=How%20much%20energy%20does%20a,2022%20was%20about%2075%20MMBtu.)

So let's do the upper limit and just multiply by 20. Obviously babies and children use less energy. So that's 1,635,320kWh or 1.64GWh.

The training of GPT-3 (175 billion parameters) in 2022 took roughly 34 days and consumed approximately 1.287 to 1.3 gigawatt-hours (GWh) of electricity. GPT-4 model, which features over a trillion parameters, required much more energy, with estimates ranging from 50 to 62.3 GWh over a 100-day training period. Training of GPT-5 is estimated to use over 1,500GWh.

So at minimum training GPT-3 cost the same as 79% of one American over 20 years. GPT-4 at minimum cost the same as 31 people. GPT-5 would be about 917 people minimum.

Now for inference

Based on data from mid-2025, a single ChatGPT query (using models like GPT-4o) consumes approximately 0.3 to 0.4 watt-hours (Wh) of electricity

The human brain consumes approximately 12 to 20 watts of power continuously. Over a 24-hour period, this translates to roughly 288 to 480 watt-hours (Wh) per day. Now to answer a question it knows is a small fraction of that. It's difficult to quantify the average energy consumption to answer a question someone already knows but let's say it takes ten seconds of brain power, so that would be 0.06Wh on inference.

So GPT uses at least five times the energy on inference compared to a human brain.

u/Erizo69 • 1 points • 2026-02-22 11:07 UTC

\>1,500GWh
that's ***wild***

u/1F61C • 1 points • 2026-02-22 11:09 UTC

Yeah, the total energy consumption of 1000+ people over 20 years. That said, the training only occurs once. It's not like a continuous cost or energy burden. Over the lifetime of the model the inference of hundreds of millions of people will greatly outweigh the training energy cost.

u/tapered_elephant • 1 points • 2026-03-11 08:30 UTC

It only occurs once... Until the next model is trained to replace it. These models have relatively short lifespans.

u/Bwint • 2 points • 2026-02-22 07:36 UTC

The inference costs are what's really important here. Say GPT-5 took the same amount of energy to train as 1,000 people would use over 20 years. Well, now that it's trained, GPT-5 can have millions of instances running with no additional training costs. At some point, the training costs

become negligible relative to the costs of running millions of instances... If the energy costs *per inference* are high relative to the energy costs of being a human, that's an important realization. I'm glad I still have at least one advantage over the clankers!

u/1F61C • 2 points • 2026-02-22 07:39 UTC

Yes, although if the energy to train was equivalent to billions or trillions of people then it wouldn't be worthwhile or at least it would take a serious amount of time to get an ROI.

I believe when we switch over to photonics that training as well as inference will become 10x to 100x cheaper. The technology is there but new so no one's taken the big financial gamble on it yet.